(1) EM Spectrum

Ranges	γ	X	UV	vis	IR	micro	radio
Energy (E)	↑						↓
Frequency (ν)	↑						↓
Wavelength (λ)	↓						↑
Wavelength (λ)	<0.01	0.01–10	10–400	400–700	700–1e6	1e6–9	>1e9

- violet · indigo · blue · green · yellow · orange · red

(2) $c = \lambda\nu$ (Speed of light = wavelength × frequency)

c	ν **(Hz)**	**kHz**	**MHz**	**GHz**
λ (**m**)	2.9979e8 (m/s)	2.9979e5 (m·kHz)	2.9979e2 (m·MHz)	2.9979e-1 (m·GHz)
nm	2.9979e17 (nm·Hz)	2.9979e14 (nm·kHz)	2.9979e11 (nm·MHz)	2.9979e8 (nm·GHz)

(3) $E = h\nu = h(c/\lambda)$ (Energy = Planck × frequency)

hc, h	λ **(m)**	λ **(nm)**	ν **(Hz)**
E (**J / photon**)	1.9864e-25 (J·m)	1.9864e-16 (J·nm)	6.626e-34 (J·s)
J / mol	1.1962e-1 (J·m/mol)	1.1962e8 (J·nm/mol)	3.9902e-10 (J·s/mol)
kJ / mol	1.1962e-4 (kJ·m/mol)	1.1962e5 (kJ·nm/mol)	3.9902e-13 (kJ·s/mol)

- Grams → mol → ×6.022e23 for photon **count**

(4) Quantum Principles

Maxwell	Light is electromagnetic radiation
Planck	Energy emitted only in fixed amounts (quanta) (black-body radiation)
Einstein	Different light has different amounts of fixed energy (photons) (photoelectric effect)
Bohr	Electrons are also quantum (line spectra); Rydberg equation
De Broglie	Electrons are particle and wave
Schrodinger	Wave equation (n, l, m_l, m_s)
Heisenberg	Position and velocity not both knowable

- $E = h\nu = h(c/\lambda) = R(1/n_f^2 - 1/n_i^2)$
- $n_i = \sqrt{1/\left(1/n_f^2 - hc/(R\lambda)\right)}$; $n_f = \sqrt{1/\left(1/n_i^2 + hc/(R\lambda)\right)}$

Variable	Value	Variable	Value
h	6.626e-34 J·s	h/R	3.0394e-16 s
c	2.9979e8 m/s	hc	1.9864e-25 J·m
R	2.18e-18 J	hc/R	9.1120e-8 m

(5) de Broglie

$$\lambda = h/(mu)$$

h	u **(m/s)**	**km/s**
m (**kg**)	6.626e-34 (J·s)	6.626e-37 (kJ·s)
g	6.626e-31 (mJ·s)	6.626e-34 (J·s)
amu	3.9892e-7 (J·s·amu/kg)	3.9892e-10 (kJ·s·amu/kg)

- 1 nm = 1e-9 m

(6) Electron Configuration

Symbol	Name	Value
n	**Principal energy level** (shell)	(1, 2, 3, ...)
l	**Sublevel** = subshell (shape)	(0, ..., n - 1)
m_l	**Orbital** (orientations)	(-l, ..., l)
m_s	**Spin** (2 per orbital)	$\pm 1/2$

- **s** (0; 0, 1/2), **p** (1; ±1, 3/6),
- **d** (2; ±2, 5/10), **f** (3; ±3, 7/14)
- 1s 2s 2p 3s 3p **4s 3d** 4p 5s 4d 5p 6s 4f 5d 1. Respect **s2 p6 d10 f14**

Trend	**→ across period**	**↓ down group**
Z_{eff}	increases	~constant
Atomic radius	decreases	increases
Ionization energy	increases	decreases
Electron affinity	more favorable	less favorable

- **Cr = [Ar]$4s^1 3d^5$ · Cu = [Ar]$4s^1 3d^{10}$** (half/full d wins) · same trick one row down: **Mo, Ag** (and Au)
- Impossible if any subshell is over-filled ($3s^3$, $2p^8$)
- Faster than writing Aufbau: read it off the table — s-block (gp 1–2) → ns, p-block (13–18) → np, d-block → $(n-1)d$; the period number is n
- Can be a transition metal; ground state only

(7) Reading an Electron Configuration

Which element?

1. Noble-gas core → its own Z: [He]2 [Ne]10 [Ar]18 [Kr]36 [Xe]54
2. Add every superscript to it = total e^- = Z (neutral atom)
3. Look up Z on the periodic table

- Cross-check: outermost subshell → period = n, block → group

Paramagnetic vs diamagnetic

1. Draw boxes for the outer subshell — **s 1 · p 3 · d 5 · f 7**
2. Fill singly, parallel spins, before pairing (Hund)
3. Count unpaired e^-

- Any unpaired → **paramagnetic** · all paired → **diamagnetic**
- Full subshells (s^2, p^6, d^{10}) are always paired

Excited vs ground state

- **Ground** = strict Aufbau order, no gaps
- **Excited** = an e^- sits higher than Aufbau requires, leaving a **gap below it**; same total e^- count

Valence electrons

- Count e^- in the **highest** n (outer s + p) = the group number, 1A→1 … 8A→8
- Everything below the highest n is **core**

(8) Ion Configuration

1. Start from the neutral atom
2. **Anion:** add e^- into the outer p until it reaches a noble-gas configuration
3. **Cation:** remove from the **highest** n **first**

- Transition metals lose **4s before 3d**
- Exam ions are main-group, not transition metals

(9) Quantum Numbers, Penetration & Shielding

- Shell n holds n^2 orbitals, $2n^2$ e^-
- **energy: s < p < d < f** (within shell, same n)
- **Pauli exclusion** — no two e^- carry the same four quantum numbers → 2 per orbital, opposite spins
- **Penetration** — orbital reaching into the core region
- ↑ shielding ⇄ ↑ reduced Z_{eff} ⇄ ↓ penetration ⇄ ↑ **energy** (reverse also holds)
- **Shielding** — only **core** (inner) e^- shield effectively; valence e^- shield each other poorly

(10) Z_{eff} & Core Electrons

$$Z_{eff} \approx Z - (\text{core } e^-)$$

1. Core e^- = total e^- − valence e^- (= the **previous noble gas**)
2. $Z_{eff} \approx Z - \text{core } e^-$

- Same period → same core, so the **rightmost** element wins
- Poor valence shielding is why Z_{eff} rises across a period

(11) Rank Atomic Size

- Down a group **bigger** (new shell) · across a period **smaller** (Z_{eff} pulls in)
- **↑ = lower-left** · **↓ = upper-right**
- Group position beats period position when both apply
- cation < neutral < anion

(12) Rank Ionization Energy

- Across a period ↑ · down a group ↓ (opposite of radius)
- ↑ **= upper-right** (He ↑ of all) · ↓ **= lower-left**
- IE_2 > IE_1 always
- **Big jump** appears once the first **core** e⁻ is removed

(13) Rank Electron Affinity

- **Most eager for an e⁻ = halogens, upper right** — this course writes it both as "most negative EA" and "largest positive EA"; either phrasing means the same element
- Noble gases and full/half-full subshells ≈ 0 or unfavorable
- EA**(X) and** IE **of X⁻ are the same magnitude** — F + e⁻ releases exactly what F⁻ → F costs

(14) Families & Diagonal Rule

1A	alkali metals	most reactive metals; ↑ **down**
2A	alkaline earth	reactive, less than 1A
7A	halogens	reactive nonmetals; ↓ **down**
8A	noble gases	**inert** — full octet

- **Diagonal rule** — an element resembles its down-right neighbor: Li~Mg · **Be~Al** · B~Si

HW-only gaps — off the exam list

Oxides — metal oxide = **basic** (K_2O, MgO, Fe_2O_3, Cr_2O_3) · nonmetal oxide = **acidic** (P_4O_{10}, SO_2, CO_2, Cl_2O_7) · Al_2O_3 and BeO = **amphoteric**

Formula from charges — 1A 1+ · 2A 2+ · 3A 3+ · 5A 3− · 6A 2− · 7A 1−; criss-cross and reduce

- Same-group swap keeps the pattern: BaS → **CaO**
- 2A + $3F_2$ → $2AF_3$ ⇒ A is 3+ (**Al**) · 3Li + Z → Li_3Z ⇒ Z is 3−, so Mg gives $\mathbf{Mg_3Z_2}$

Bond-type ID — ionic **and** covalent together = a salt of a **polyatomic ion** (NH_4NO_3, K_2SO_4); all-nonmetal = covalent only

Family reactions

- 2K(s) + $2H_2O$(l) → 2KOH(aq) + H_2(g)
- H_2(g) + Cl_2(g) → 2HCl(g)

Odds & ends

- Lanthanides = 4f row (**Ce**) · actinides = 5f (U)
- Diatomic: H_2 N_2 O_2 F_2 Cl_2 Br_2 I_2
- Metallic character ↑ down, ↓ across → **lower-left** wins
- Isoelectronic = matches a noble gas e⁻ count (K^+, Cl^-, Ca^{2+} = Ar)

(15) Lattice Energy

Energy **released** when gaseous ions form the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

1. Compare charges first
2. If charges tie, smaller ions win

- ↑ charge → ↑ **U** (dominant) · ↑ ionic radius → ↓ U

(16) Electronegativity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

- Read EN off the chart — EN ↑ up and to the right, F highest, noble gases excluded

(17) Bond Polarity

1. Take the absolute difference (ΔEN)
2. Classify

~0	nonpolar covalent
0.4 – 1.9	polar covalent
≥ 2.0	ionic

- **"Most polar"** → ΔEN **alone** (largest difference wins)
- **"Strongest / shortest bond" → bond order FIRST:** triple > double > single. ΔEN only breaks ties between equal orders — a C≡N beats a high-ΔEN Si=O
- Larger ΔEN → stronger, shorter bond, but the guide flags this as *not always true*

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \#\,\text{bonds}]$$

1. FC on every atom
2. Check ΣFC = the overall charge
3. Pick the structure with FCs **nearest 0**
4. Put any −1 on the **more electronegative** atom

- Equivalent resonance forms → real bonds are the **average** (e.g. 1½)

(19) Lewis Structures ×4

1. Total valence e⁻ — **+**1 per negative charge, **−**1 per positive
2. Central atom = least electronegative, never H
3. Single bonds out to every atom
4. Fill outer octets with lone pairs (H takes 2)
5. Leftover e⁻ onto the central atom
6. Central below 8 → pull an outer lone pair into a double/triple bond, **unless** Al/B/Be
7. Check formal charges — see (18)

Four structures — one of each type:

(a) Normal octet	8 valence e⁻ per atom (H wants 2) — H_2O, CH_4, NH_3, CO_2, CH_2Cl_2, CH_3CH_2Cl
(b) Incomplete octet	**Al, B, Be only** — content below 8, do not add double bonds to fix. BF_3/BCl_3/AlI_3 leave 6 e⁻ on the center; $BeCl_2$ leaves 4
(c) Expanded octet	Central atom **period 3 or below** (S, P, Cl, Xe, As, I) uses empty d orbitals for >8 — $AsCl_5$, XeI_2, SO_4^{2-}
(d) Organic	**–OH** alcohol · **C=O** aldehyde/ketone · **–COOH** carboxylic acid · $\mathbf{–NH_2}$ amine

- Every C makes 4 bonds; a –COOH carbon carries one **=O** and one **–O–H**; every O gets 2 lone pairs
- Ions go in brackets with the charge outside
- If the all-single-bond form leaves the center with a nonzero FC, add double bonds to zero it (e.g. two S=O in SO_4^{2-})

(20) Photoelectric Effect ($h\nu$ - binding / work)

$$KE = h\nu - \Phi = h(c/\lambda) - \Phi = \frac{1}{2} m_e u^2$$

$$u = \sqrt{(2hc/m_e)(1/\lambda - 1/\lambda_{thr})}$$

Variable	Value
h	6.626e-34 J·s
m_e	9.11e-31 kg
hc	1.9864e-25 J·m
$2hc/m_e$	4.3609e5 m³/s²

Have	Want	Use
ν_{thr} or λ_{thr}	Φ	$\Phi = h\nu_{thr} = h(c/\lambda_{thr})$
Φ	λ_{thr} (max ej. λ)	$\lambda_{thr} = hc/\Phi$
Φ	Φ per mol	Φ × N_A (/1e3 for kJ/mol)
Φ, λ	u	1. $KE = h(c/\lambda) - \Phi$ 2. $u = \sqrt{2\,KE/m_e}$
λ, λ_{thr}	u	2nd eq above (skips Φ **and** KE)

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}, \quad \Delta p = m \Delta u$$

1. $\Delta p = m \cdot \Delta u$ (kg, m/s)
2. $\Delta x \geq h/(4\pi \cdot \Delta p)$

- Result is the **minimum** uncertainty
- Rearrange the same way to solve for Δu
- Keep everything in kg, m, s

(22) Born–Haber Cycle

$$\Delta H^\circ{}_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2} BE + EA + U$$

1. List every step with its sign: sublimation (+), IE (+), bond dissociation (+), EA (usually −), U (−)
2. **Halve** the bond dissociation if you need only one atom
3. Chain: elements → gaseous atoms → gaseous ions → solid
4. Set the chain equal to $\Delta H^\circ{}_f$
5. Solve for the single unknown

- Use IE_2 as well when the cation is 2+
- U runs ions(g) → solid, so it comes out **negative**; report |U| if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma \Delta H(\text{bonds broken}) - \Sigma \Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients
2. Σ BE(broken), always positive
3. Σ BE(formed), subtracted
4. ΔH = Σ broken − Σ formed

- Bonds unchanged on both sides cancel — skip them
- Negative ΔH = exothermic
- For "heat released by *g* of X," convert g → mol and scale